Supplementary evidence table

Hierarchical Taxonomic Abstraction for the Safe Handling of Novel Objects in Autonomous Driving Perception (v1, doi:10.5281/zenodo.21593472)

Accompanies the review exchange. Per-detection data: `cases.csv` (54 rows). This sheet reports aggregates with explicit denominators.

Evaluation set. SegmentMeIfYouCan Road Anomaly, 40 images, 54 detections. Detector: YOLOv8s, class-agnostic NMS, confidence ≥ 0.20. CLIP: ViT-B/32. No training or fine-tuning.

Population split (novelty is a proxy). A detection is labelled *out-of-taxonomy* when the detector's COCO class has no node in our taxonomy (e.g. giraffe, airplane). The detector already recognizes these objects; this is a *label-space gap*, not open-world novelty. It is distinct from genuine, scale-dependent undecidability (the "shampoo-bottle" regime), which is not tested here. Out-of-taxonomy $n = 20$; in-taxonomy $n = 34$.

Thresholds. Leaf-softmax temperature $\tau = 0.06$; commit-mass $\theta = 0.40$ (minimum subtree probability mass to descend one level); minimum best-leaf cosine $\sigma = 0.20$ (below this: immediate UNKNOWN). Safety floors: Living Being, Vehicle, Static Object.

Population	Metric	Flat (argmax)	Flat (reject@0.5)	**Hierarchical**
Out-of-taxonomy ($n = 20$)	**safe-handled**	0/20	0/20	**20/20**
	(abstracted / UNKNOWN)	—	—	9 / 11
	of which *plausibly* abstracted/flagged	—	—	15/20
	importance-weighted safe	0%	0%	**100%**
	confident wrong leaf	20/20	0/20	0/20
	dropped (missed)	0/20	20/20	0/20
In-taxonomy ($n = 34$)	**useful category**	16/34	—	**26/34**
	honest UNKNOWN	—	—	8/34
	wrong branch	—	—	0/34

Definitions. *Safe-handled* (out-of-taxonomy): the detection is neither given a specific leaf nor dropped; it is abstracted to a category at or below a safety floor, or flagged UNKNOWN. *Useful category* (in-taxonomy): outcome is not UNKNOWN, the reported node is below a floor and on the correct root-to-leaf branch. The flat baseline's 0/20 is defined by the comparison rule (arg-max can only emit a leaf; the reject variant drops), not yet by a fully equivalent empirical open-set baseline.

Plausibility (stricter than safe-handled). *Safe-handled* only requires that a detection is not given a confident leaf and not dropped. A stricter check, *plausibly abstracted / flagged*, additionally requires the coarse category to be semantically correct: 15/20 pass (11 flagged UNKNOWN, 4 abstracted to the correct floor), while 5 abstract to an implausible branch (airplane, kite and frisbee to *Living Being*). Those are out-of-ODD classes with no correct branch in the taxonomy, which is precisely the gap a genuine open-world novelty test must target. See the `plausible` column in `cases.csv`.

Scope and caveats. (i) Novelty is a label-space proxy, not open-world novelty. (ii) No box-level ground truth: object-level false negatives and missed objects are outside the evaluation; the KPI scores only detected objects. (iii) Sensor degradation and broader domain shift are not evaluated. (iv) The taxonomy provides bounded semantic *evidence*; authorization of any manoeuvre is a separate (execution-governance) decision.